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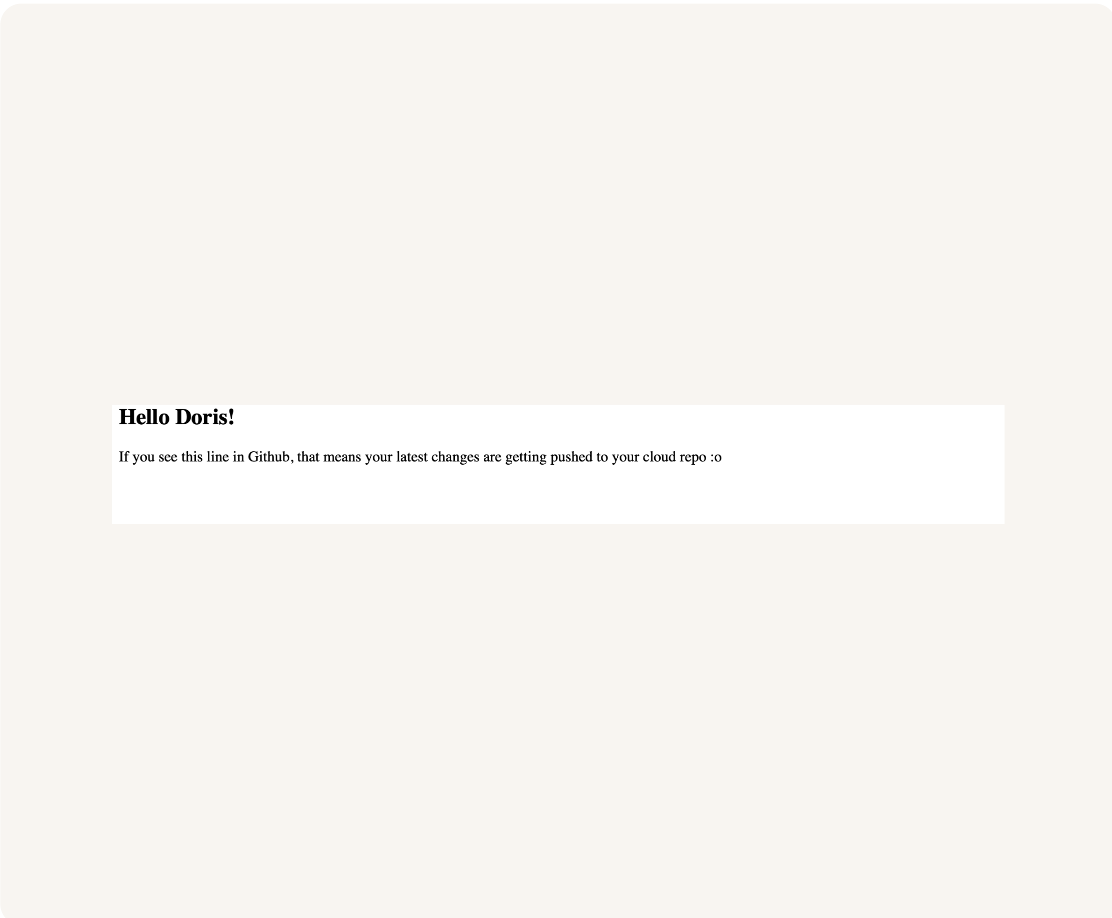
Deploy a Web App with CodeDeploy

DE delorezdoris@gmail.com



Introducing Today's Project!

In this project, I will demonstrate how to launch a deployment environment using AWS CloudFormation,write deployment scripts to automate deployment commands and Implement a disaster recovery technique - roll back a deployment!

Key tools and concepts

Services I used were code deploy,cloud formation,codebuild,code connection,codeartifact etc. Key concepts I learnt include deployment,deployment groups,scripts and the importance of rebuilding your project before you deploy.

Project reflection

This project took me approximately 3 hours and 30 minutes. The most challenging part was the seperation of the deployment instance from the production instance . It was most rewarding to see the deployed webapp.

This project is part five of a series of DevOps projects where I'm building a CI/CD pipeline! I'll be working on the next project tomorrow.

DE delorezdoris@gmail.com
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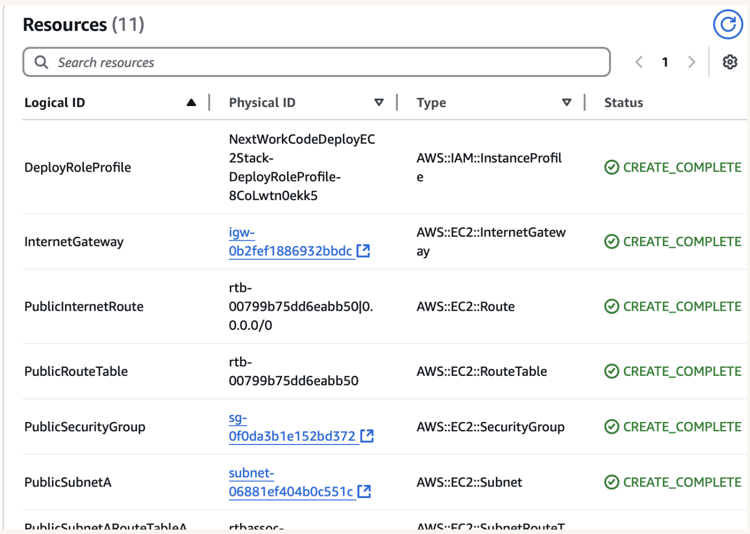
Deployment Environment

To set up for CodeDeploy, I launched an EC2 instance and VPC because they will make up the production environment where we deploy the webapp.

Instead of launching these resources manually, I used cloud formation. When I need to delete these resources created by the cloud formation template I can easily delete the entire stack.

Other resources created in this template include networking resources like VPC,subnet,internet gateway etc. They're also in the template because a production environment has specific traffic requirements to be enabled.



Resources (11)

Logical ID	Physical ID	Type	Status
DeployRoleProfile	NextWorkCodeDeployEC2Stack-DeployRoleProfile-8CoLwtn0ekk5	AWS::IAM::InstanceProfile	CREATE_COMPLETE
InternetGateway	igw-0b2fef1886932bbdc	AWS::EC2::InternetGateway	CREATE_COMPLETE
PublicInternetRoute	rtb-00799b75dd6eabb50|0.0.0.0/0	AWS::EC2::Route	CREATE_COMPLETE
PublicRouteTable	rtb-00799b75dd6eabb50	AWS::EC2::RouteTable	CREATE_COMPLETE
PublicSecurityGroup	sg-0f0da3b1e152bd372	AWS::EC2::SecurityGroup	CREATE_COMPLETE
PublicSubnetA	subnet-06881ef404b0c551c	AWS::EC2::Subnet	CREATE_COMPLETE
PublicSubnetARouteTableA	rtbassoc-	AWS::EC2::SubnetRouteT	

Deployment Scripts

Scripts are like mini-programs that automate tasks . To set up CodeDeploy, I also wrote scripts to automate the deployment tasks.

install_dependencies will set up all the software needed to run our website by installing programs (called Tomcat and Apache) that handle internet traffic and host our application

start_server.sh will start both Tomcat (our Java application server) and Apache (our web server) and makes sure they'll restart automatically if the EC2 instance ever reboots.

stop_server.sh will safely stop web server services by first checking if they're running. It uses pgrep to check for running processes of Apache (httpd) and Tomcat, and only attempts to stop the services if they are actually active.

DE
delorezdoris@gmail.com
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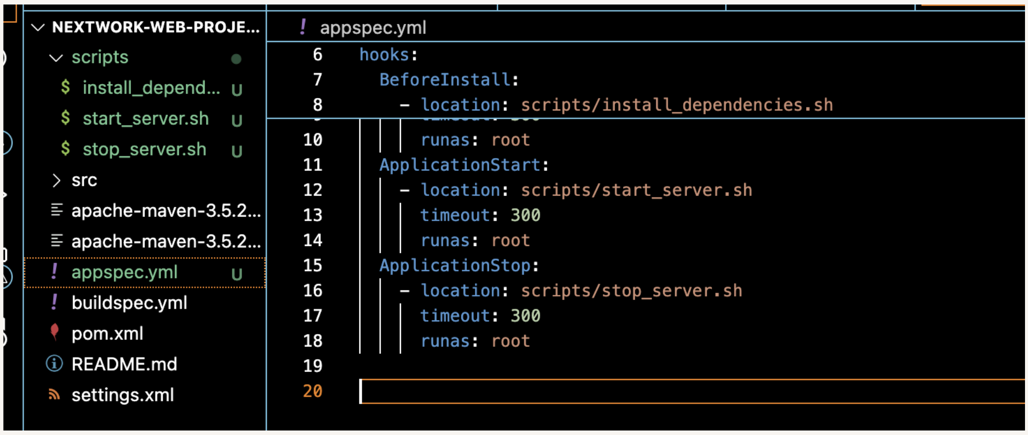
appspec.yml

Then, I wrote an appspec.yml file which is the instruction manual for CodeDeploy. The key sections in appspec.yml are source: /target/nextwork-web-project.war: "Take this WAR file from our artifact' destination: /usr/share/tomcat10/webapps/: ".

I also updated buildspec.yml because it's telling CodeBuild: "After you build the app, make sure to include these additional files in the final package." CodeDeploy will need the appspec.yml to properly deploy the application.

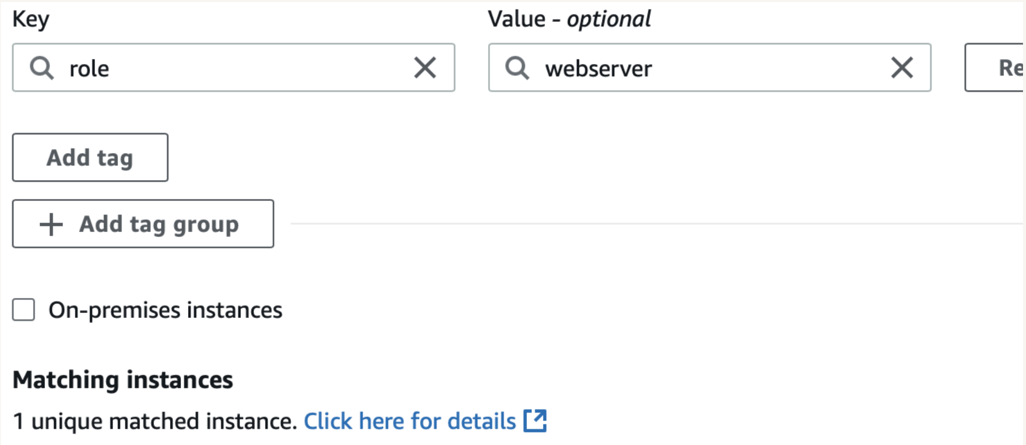


Setting Up CodeDeploy

A deployment group is s a collection of EC2 instances that are grouped to deploy something together. A CodeDeploy application is like the main folder for your deployment project. It doesn't do much on its own, but it helps you organize everything

To set up a deployment group, you also need to create an IAM role to get permissions to access and manage AWS resources on your behalf. It allows CodeDeploy to access EC2 instances to deploy applications,Read application artifacts from S3 buckets etc

Tags are helpful as CodeDeploy can identify the target EC2 instances for deployment. I used the tag to make sure it finds and deploy to the correct instance.


Key
role
Value - optional
webserver
Re
Add tag
Add tag group
On-premises instances
Matching instances
1 unique matched instance. Click here for details

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delorezdoris@gmail.com
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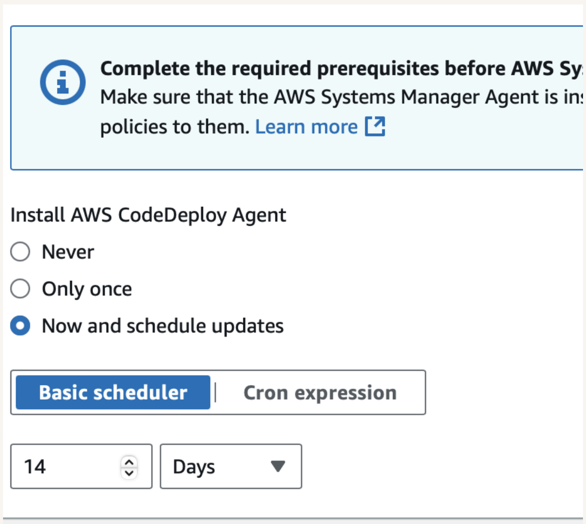
Deployment configurations

Another key settings is the deployment configuration, which affects how quickly you'd like to roll out your application.I used CodeDeployDefault.AllAtOnce,s which deploys the application to all instances in the deployment group at the same time.

In order for the EC2 instance to communicate with CodeDeploy, a CodeDeploy Agent is set up .Whenever you initiate a deployment, it's the CodeDeploy Agent that receives the deployment instructions from CodeDeploy and carries them out on the instance.

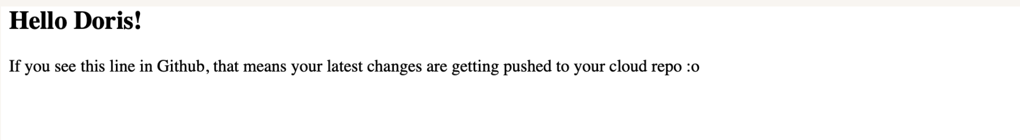


Success!

A CodeDeploy deployment represents a single update to your application, with its own unique ID and history. When you create a deployment, you're telling CodeDeploy: Which version of the application to deploy, where to deploy and how to deploy it.

I had to configure a revision location, which means the place where CodeDeploy looks to find your application's build artifacts. My revision location was my s3 buckets

To check that the deployment was a success, I visited my public ipv4 dns address and I saw my application working.



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